



Bank Customer Churn Analysis

Project Overview

You are working as a **Data Analyst** at a leading bank.

The management is concerned about **customers leaving the bank (churn)** and wants insights to improve retention.

Your Task:

- Clean and model the data
- Apply DAX calculations (hints provided)
- Build interactive dashboards
- Generate actionable business insights

Dataset Description

Dataset Customer Churn:- [Customer Churn Dataset](#)

Table Name	Description
CustomerInfo	Customer IDs and surnames
BankChurn	Main dataset with demographics, balances, tenure, etc.

Geography	Country details
Gender	Gender mapping
CreditCard	Credit card ownership
ActiveCustomer	Customer activity status
ExitCustomer	Customer exit/retention status

Data Model (Star Schema)

Fact Table:

- BankChurn

Dimension Tables:

- CustomerInfo, Geography, Gender, CreditCard, ActiveCustomer, ExitCustomer

Relationships:

- Connect ID columns from dimension tables to BankChurn
- Direction: One-to-Many

1 Data Cleaning

- Remove duplicate rows
- Fix missing/null values
- Standardize column names
- Validate date formats and numeric columns
- Ensure IDs are consistent

2. Data Modeling

- Implement **Star Schema**
- Hide ID columns from report view
- Set lookup tables for filtering

3. DAX

Calculated Columns

1. **Customer Age Group** – Categorize as Young, Middle Age, Senior
Hint: Use `IF()` with logical conditions.
2. **Tenure Category** – Categorize as New or Loyal
Hint: Compare tenure values using `IF()`.
3. **Balance Status** – Identify zero balance vs has balance
Hint: Use `IF()` to compare Balance.

Measures

1. **Total Customers** – Count total rows
Hint: Use counting function.

2. **Exited Customers** – Count only exited customers
Hint: Use conditional calculation with filter.
3. **Retention Rate (%)** – Percentage of retained customers
Hint: Use division of retained customers by total.
4. **Average Balance** – Average of customer balances
Hint: Use average function.
5. **Average Salary** – Average estimated salary
Hint: Use average function.

Phase 4 – Dashboard Design

Page 1 – Customer Overview

- **Cards:** Total Customers, Retention Rate, Active Member %
- **Donut Chart:** Gender distribution
- **Bar Chart:** Customers by Geography
- **Tree Map:** Age group distribution

Page 2 – Churn Analysis

- **Clustered Bar Chart:** Exited vs Retained
- **Stacked Column Chart:** Churn by Geography & Gender
- **Funnel Chart:** Active → Inactive → Exit
- **Line Chart:** Average Balance vs Age

Page 3 – Product & Credit Analysis

- **Pie Chart:** Credit card holders vs non-holders
- **Column Chart:** Exit rate by Number of Products

- **Matrix:** Geography vs Exit Rate
- **Gauge Chart:** Retention Rate (%)